

Reviewer Pack — Minimal Automorphism Group Rank of Lie Algebras vs. Monotone...

Entry 95b22936e411 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Automorphism Group Rank of Lie Algebras vs. Monotone Circuit Lower Bounds for k-CLIQUE

Entry ID: 95b22936e411 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 12:20:05 UTC

1. Conjecture

Field A (mathematical branch): Lie Algebra Theory **Field B** (complexity object): Monotone Circuit Complexity for k-CLIQUE

Statement:

{‘text’: ‘For every n , there exists a polynomial $f(n)$ such that the minimal rank of any Lie algebra automorphism group corresponding to an n -input k-CLIQUE monotone circuit is at least $f(n)$.’, ‘quantitative_relation’: ‘ $E[X(\text{automorphism_group_rank})] = \Theta(f(n))$ ’, ‘falsifier’: ‘An n -input k-CLIQUE monotone circuit with an automorphism group rank less than a polynomially bounded value that holds for all sufficiently large n .’}

Rationale (proposer’s reasoning):

{‘text’: ‘Lie algebras have been used in the study of symmetry in mathematics and physics. The minimal automorphism group rank measures the complexity of the symmetry structure of a Lie algebra. A direct correspondence between this rank and the complexity of monotone circuits for k-CLIQUE could reveal new insights into the hardness of k-CLIQUE problems.’, ‘explanation’: ‘This conjecture proposes that there is a connection between the abstract concept of symmetry in Lie algebras and the computational complexity of certain circuit-based problems. If true, it would be a novel application of Lie algebra theory to complexity theory.’}

Taxonomy category: LieAlgebra_to_MonotoneCircuit (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): eaa970b152fe1059

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal automorphism group rank of k-CLIQUE monotone circuits is measured against a polynomially bounded threshold, where the mean rank across all seeds does not exceed the polynomial $f(n)$ for $n \geq 40$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Automorphism Group Rank Lie Algebras AND Monotone Circuit Lower Bounds k-CLIQUE - Lie Algebra Theory AND k-CLIQUE Monotone Circuit Complexity automorphism group rank - Polynomial Bounds Minimal Rank Automorphism Groups k-CLIQUE Monotone Circuits

Top relevant hits considered: - [http://arxiv.org/abs/2101.11518v4] Classification problem of simple Hom-Lie algebras - [http://arxiv.org/abs/1405.2700v1] Zero Excess and Minimal Length in Finite Coxeter Groups - [http://arxiv.org/abs/2302.10064v3] On minimal easily computable dimension group algebras, and group codes - [http://arxiv.org/abs/1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [http://arxiv.org/abs/math/0403509v5] Leibniz algebras, Lie racks, and digroups - [http://arxiv.org/abs/2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [http://arxiv.org/abs/1511.09051v3] On automorphism groups of affine surfaces - [http://arxiv.org/abs/2104.14760v4] Automorphism and outer automorphism groups of Right-angled Artin groups are not relatively hyperbolic

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_k_clique_circuit(n):
        # Generate a random k-clique circuit (simplified version for testing)
        return [random.randint(0, n-1) for _ in range(random.randint(2, 5))]

    def compute_automorphism_group_rank(circuit):
        # Simplified computation of automorphism group rank
        return len(set(circuit))

    n_values = [5, 10, 15, 20, 30, 40]
    ranks = []
    for n in n_values:
        for _ in range(5): # Test with 5 instances per n
            circuit = generate_k_clique_circuit(n)
            rank = compute_automorphism_group_rank(circuit)
```

```

        ranks.append((n, rank))

mean_rank = sum(rank for _, rank in ranks) / len(ranks)
f_n = 1099511627776 # Example polynomial function for testing

return {
    "metric_name": "Automorphism Group Rank",
    "metric_value": mean_rank,
    "instances_tested": len(ranks),
    "conjecture_holds": mean_rank >= f_n,
    "counterexample": "" if mean_rank >= f_n else f"n=40, rank={mean_rank}, f(n)={f_n}"
}

if __name__ == "__main__":
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((result["seed"] for result in results if not result["conjecture_holds"]))
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.2666666666666666, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.4333333333333333, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.3333333333333335, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.1333333333333333, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.2333333333333334, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 2.9333333333333333, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 2.8666666666666667, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.3, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.3333333333333335, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}
TRIAL: {'metric_name': 'Automorphism Group Rank', 'metric_value': 3.2333333333333334, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'n=40, rank=3.3666666666666667, f(n)=1099511627776'}

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The conjecture was falsified by a counterexample where an $n=40$ k-CLIQUE monotone circuit has an automorphism group rank of 3.2333333333333333, which i | next: Investigate further to determine if there are other counterexamples or if this is a general trend for larger values of n .

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15819
2	preregistration	ollama_remote	glm4:latest	0	0	11325
3	novelty	ollama_remote	glm4:latest	0	0	8203
4	novelty	ollama_remote	glm4:latest	0	0	9986
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	25298
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8051
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7702
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7892
9	judge	ollama_remote	glm4:latest	0	0	14385

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 108661 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/95b22936e411.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/95b22936e411.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/95b22936e411.tar.gz` (if generated)